1. Data Cleansing Steps

In a single query, perform the following operations and generate a new table in the `data_mart` schema named `clean_weekly_sales`:

- Convert the `week_date` to a `DATE` format
- Add a `week_number` as the second column for each `week_date` value, for example any value from the 1st of January to 7th of January will be 1, 8th to 14th will be 2 etc
- Add a `month_number` with the calendar month for each `week_date` value as the 3rd column
- Add a `calendar_year` column as the 4th column containing either 2018, 2019 or 2020 values
- Add a new column called `age_band` after the original `segment` column using the following mapping on the number inside the `segment` value

segment	**age_band**
1	Young Adults
2	Middle Aged
3 or 4	Retirees

- Add a new `demographic` column using the following mapping for the first letter in the `segment` values:

segment	demographic
C	Couples
F	Families

- Ensure all `null` string values with an `"unknown"` string value in the original `segment` column as well as the new `age_band` and `demographic` columns
- Generate a new `avg_transaction` column as the `sales` value divided by `transactions` rounded to 2 decimal places for each record

Answer :

```sql
select try_convert(DATE,week_date,3) as week_date
,DATEPART(week,try_convert(DATE,week_date,3)) as week_number
,DATEPART(month,try_convert(DATE,week_date,3)) as month_number
,DATEPART(year,try_convert(DATE,week_date,3)) as calendar_year
,CASE
    when RIGHT(segment,1) = '1' then 'Young Adults'
    when RIGHT(segment,1) = '1' then 'Middle Aged'
    when RIGHT(segment,1) in ('3','4') then 'Retirees'
    else 'unknown'
END as age_band
,CASE
    when LEFT(segment,1) = 'C' then 'Couples'
    when LEFT(segment,1) = 'F' then 'Families'
    else 'unknown'
END as demographic
,ROUND((CAST(sales as decimal)/transactions),2) as avg_transaction ,region,platform,segment,customer_type,transactions,sales
into weekly_salesss
from weekly_sales;
```

	week_date	week_number	month_number	calendar_year	age_band	demographic	avg_transaction	region	platform	segment	customer_type	transactions	sales
1	2018-09-03	36	9	2018	Retirees	Couples	246.1400000000	EUROPE	Shopify	C3	New	7	1723
2	2018-09-03	36	9	2018	Young Adults	Couples	28.9700000000	SOUTH AMERICA	Retail	C1	Existing	2240	64890
3	2018-09-03	36	9	2018	Retirees	Families	42.1700000000	EUROPE	Retail	F3	New	1667	70303
4	2018-09-03	36	9	2018	Retirees	Couples	53.1300000000	AFRICA	Retail	C4	Existing	91442	4858213
5	2018-09-03	36	9	2018	unknown	Couples	44.1200000000	SOUTH AMERICA	Retail	C2	Existing	595	26250
6	2018-09-03	36	9	2018	unknown	unknown	50.2200000000	CANADA	Retail	null	Existing	6943	348668
7	2018-09-03	36	9	2018	Retirees	Couples	174.5000000000	OCEANIA	Shopify	C3	New	361	62994
8	2018-09-03	36	9	2018	unknown	Couples	209.2900000000	USA	Shopify	C2	Existing	942	197148
9	2018-09-03	36	9	2018	Retirees	Couples	178.0700000000	AFRICA	Shopify	C3	New	211	37572
10	2018-09-03	36	9	2018	Retirees	Couples	157.4800000000	CANADA	Shopify	C4	Existing	58	9134
11	2018-09-03	36	9	2018	Young Adults	Couples	32.2500000000	EUROPE	Retail	C1	New	4189	135087
12	2018-09-03	36	9	2018	unknown	unknown	40.1900000000	SOUTH AMERICA	Retail	null	Guest	401420	16132084
13	2018-09-03	36	9	2018	unknown	Families	159.1600000000	ASIA	Shopify	F2	New	156	24829
14	2018-09-03	36	9	2018	unknown	Families	37.1000000000	AFRICA	Retail	F2	New	62896	2333272
15	2018-09-03	36	9	2018	unknown	unknown	182.0400000000	OCEANIA	Shopify	null	Guest	5604	1020147
16	2018-09-03	36	9	2018	Retirees	Families	317.8500000000	EUROPE	Shopify	F3	Existing	127	40367
17	2018-09-03	36	9	2018	Young Adults	Couples	43.6600000000	CANADA	Retail	C1	Existing	51833	2262960
18	2018-09-03	36	9	2018	Retirees	Families	158.2600000000	ASIA	Shopify	F3	New	81	12819

2. Data Exploration

```sql
--1. What day of the week is used for each week_date value?

select DISTINCT DATENAME(WEEKDAY,week_date) as DayName
FROM weekly_salesss;
```

	DayName
1	Monday

```sql
--2. What range of week numbers are missing from the dataset?

select * from generate_series(1,52)
EXCEPT
select distinct DATEPART(week,week_date) as value from weekly_salesss;
```

	value
11	11
12	12
13	37
14	38
15	39
16	40
17	41
18	42
19	43
20	44
21	45
22	46
23	47
24	48
25	49
26	50
27	51
28	52

```sql
--3. How many total transactions were there for each year in the dataset?

select calendar_year,SUM(CAST(transactions as bigint)) as total_transactions
from weekly_salesss
group by calendar_year;
```

	calendar_year	total_transactions
1	2018	3464064600
2	2019	3656392850
3	2020	3756930200

```sql
--4. What is the total sales for each region for each month?

select month_number,region,sum(CAST(sales as bigint)) as total_sales
from weekly_salesss
group by month_number,region
order by month_number,region;
```

	month_number	region	total_sales
1	3	AFRICA	5677674800
2	3	ASIA	5297707930
3	3	CANADA	1446343290
4	3	EUROPE	353370930
5	3	OCEANIA	7832828880
6	3	SOUTH AMERICA	710231090
7	3	USA	2253530430
8	4	AFRICA	19117835040
9	4	ASIA	18046287070
10	4	CANADA	4845525940
11	4	EUROPE	1273342550
12	4	OCEANIA	25997676200
13	4	SOUTH AMERICA	2384515310
14	4	USA	7597863230
15	5	AFRICA	16472447380
16	5	ASIA	15262853990
17	5	CANADA	4123783650
18	5	EUROPE	1093383890
19	5	OCEANIA	22156573040
20	5	SOUTH AMERICA	2013918090
21	5	USA	6559671210
22	6	AFRICA	17675597600
23	6	ASIA	16194828890
24	6	CANADA	4438466980
25	6	EUROPE	1228138260
26	6	OCEANIA	23718847440

Query executed successfully. | SF-CPU-482\MSSQLSERVER01 (1... | SF-CPU-482\install (85) | data_mart | 00:00:00 | Row: 1, Col: 1 | 49 rows

```sql
--5. What is the total count of transactions for each platform

select platform,sum(CAST(transactions as bigint)) as total_transactions
from weekly_salesss
group by platform;
```

	platform	total_transactions
1	Shopify	59251690
2	Retail	10818135960

```sql
--6. What is the percentage of sales for Retail vs Shopify for each month?

with sales_monthly as (
    select month_number,sum(CAST(sales as bigint)) as total_sales_monthly
    from weekly_salesss
    group by month_number
)
,sales_monthly_platform as (
    select month_number,platform,sum(CAST(sales as bigint)) as sales_month_platform
    from weekly_salesss
    group by month_number,platform
)
select *,100 * ROUND(CAST(sales_month_platform as decimal)/total_sales_monthly,4)
from sales_monthly_platform smp
left join sales_monthly sm
on smp.month_number = sm.month_number
order by smp.month_number;
```

	month_number	platform	sales_month_platform	month_number	total_sales_monthly	(No column name)
1	3	Retail	22991884170	3	23571687350	97.5400000000000000
2	3	Shopify	579803180	3	23571687350	2.4600000000000000
3	4	Retail	77355922340	4	79263045340	97.5900000000000000
4	4	Shopify	1907123000	4	79263045340	2.4100000000000000
5	5	Shopify	1824249020	5	67682631250	2.7000000000000000
6	5	Retail	65858382230	5	67682631250	97.3000000000000000
7	6	Retail	70499492600	6	72477143620	97.2700000000000000
8	6	Shopify	1977651020	6	72477143620	2.7300000000000000
9	7	Shopify	2142393610	7	79023308090	2.7100000000000000
10	7	Retail	76880914480	7	79023308090	97.2900000000000000
11	8	Retail	71877938350	8	74039198440	97.0800000000000000
12	8	Shopify	2161260090	8	74039198440	2.9200000000000000
13	9	Retail	11045068570	9	11342766550	97.3800000000000000
14	9	Shopify	297697980	9	11342766550	2.6200000000000000

```sql
--7. What is the percentage of sales by demographic for each year in the dataset?

with sales_year as (
    select calendar_year, sum(CAST(sales as bigint)) as sales_by_year
    from weekly_salesss
    group by calendar_year
)
,sales_year_demo as (
    select calendar_year,demographic, sum(CAST(sales as bigint)) as sales_by_year_demographic
    from weekly_salesss
    group by calendar_year,demographic
)
select *,ROUND(100 * CAST(sales_by_year_demographic as decimal)/sales_by_year,2) as percentage_of_sales
from sales_year_demo syd
left join sales_year sy
on syd.calendar_year = sy.calendar_year
order by syd.calendar_year;
```

	calendar_year	demographic	sales_by_year_demographic	calendar_year	sales_by_year	percentage_of_sales
1	2018	unknown	53694341060	2018	128973808270	41.630000000000000
2	2018	Families	41255580330	2018	128973808270	31.990000000000000
3	2018	Couples	34023886880	2018	128973808270	26.380000000000000
4	2019	Couples	37492519350	2019	137460325000	27.280000000000000
5	2019	Families	44639183440	2019	137460325000	32.470000000000000
6	2019	unknown	55328622210	2019	137460325000	40.250000000000000
7	2020	Couples	40459107650	2020	140965647370	28.700000000000000
8	2020	unknown	54363159070	2020	140965647370	38.560000000000000
9	2020	Families	46143380650	2020	140965647370	32.730000000000000

```sql
--8. Which age_band and demographic values contribute the most to Retail sales?

with percent_sales as (
    select age_band, demographic,SUM(CAST(sales as bigint)) as total_sales_ageBand_and_demographic_wise,
        ( 100*SUM(CAST(sales as decimal))/SUM(SUM(CAST(sales as bigint))) OVER() ) as percentage_of_sales
    from weekly_salesss
    where platform = 'Retail'
    group by age_band, demographic
)
select *
from percent_sales
order by percentage_of_sales DESC;
```

	age_band	demographic	total_sales_ageBand_and_demographic_wise	percentage_of_sales
1	unknown	unknown	160672855330	40.521806
2	Retirees	Families	66346869160	16.732726
3	Retirees	Couples	63669238510	16.057426
4	unknown	Families	43540915540	10.981049
5	Young Adults	Couples	26029227970	6.564589
6	unknown	Couples	18541603300	4.676205
7	Young Adults	Families	17708892930	4.466195

Before & After Analysis

This technique is usually used when we inspect an important event and want to inspect the impact before and after a certain point in time.

Taking the `week_date` value of `2020-06-15` as the baseline week where the Data Mart sustainable packaging changes came into effect.

We would include all `week_date` values for `2020-06-15` as the start of the period **after** the change and the previous `week_date` values would be **before**

Using this analysis approach - answer the following questions:

1. What is the total sales for the 4 weeks before and after `2020-06-15`? What is the growth or reduction rate in actual values and percentage of sales?

```sql
with sales_breakup as (
    select
    SUM(
        CASE
            when week_date between DATEADD(WEEK,-4,'2020-06-15') and DATEADD(DAY,-1,'2020-06-15') then CAST(sales as bigint)
            else 0
        END
       ) as total_sales_4_weeks_before_sustainable_packaging,
    SUM(
        CASE
            when week_date between '2020-06-15' and DATEADD(WEEK,3,'2020-06-15') then CAST(sales as bigint)
            else 0
        END
       ) as total_sales_4_weeks_after_sustainable_packaging
    from weekly_salesss
)
select (total_sales_4_weeks_after_sustainable_packaging - total_sales_4_weeks_before_sustainable_packaging) as actual_sales_difference,
    100*(CAST(total_sales_4_weeks_after_sustainable_packaging as decimal) - total_sales_4_weeks_before_sustainable_packaging)/total_sales_4_weeks_before_sustainable_packaging
        as percentage_change
from sales_breakup;
```

	actual_sales_difference	percentage_change
1	-268841880	-1.14601799022437

I mean the numbers are bad but not that bad –1.15 percentage change in sales when introduced the sustainable packaging this is initial sign that users don’t concern themselves with sustainable packaging.

2. What about the entire 12 weeks before and after?

```sql
--2. What about the entire 12 weeks before and after?

with sales_breakup as (
    select
    SUM(
        CASE
            when week_date between DATEADD(WEEK,-12,'2020-06-15') and DATEADD(DAY,-1,'2020-06-15') then CAST(sales as bigint)
            else 0
        END
       ) as total_sales_12_weeks_before_sustainable_packaging,
    SUM(
        CASE
            when week_date between '2020-06-15' and DATEADD(WEEK,11,'2020-06-15') then CAST(sales as bigint)
            else 0
        END
       ) as total_sales_12_weeks_after_sustainable_packaging
    from weekly_salesss
)
select (total_sales_12_weeks_after_sustainable_packaging - total_sales_12_weeks_before_sustainable_packaging) as actual_sales_difference,
    100*(CAST(total_sales_12_weeks_after_sustainable_packaging as decimal) - total_sales_12_weeks_before_sustainable_packaging)/total_sales_12_weeks_before_sustainable_packaging
        as percentage_change
from sales_breakup;
```

	actual_sales_difference	percentage_change
1	-1559815570	-2.18882372009084

Again, even after 12 weeks the end user doesn't concern themselves with sustainability i guess the marketing department should consider marketing sustainable packaging as the next big thing like how apple did it, apple doesn't advertise sustainable packaging even though they use it they advertise their company as **carbon neutral** making the keywords resonate more with end users.

3. How do the sale metrics for these 2 periods before and after compare with the previous years in 2018 and 2019?

2019

```sql
with sales_breakup as (
    select
    SUM(
        CASE
            when week_date between DATEADD(WEEK,-12,'2019-06-15') and DATEADD(DAY,-1,'2019-06-15') then CAST(sales as bigint)
            else 0
        END
       ) as total_sales_12_weeks_before_sustainable_packaging,
    SUM(
        CASE
            when week_date between '2019-06-15' and DATEADD(WEEK,11,'2019-06-15') then CAST(sales as bigint)
            else 0
        END
       ) as total_sales_12_weeks_after_sustainable_packaging
    from weekly_salesss
)
select (total_sales_12_weeks_after_sustainable_packaging - total_sales_12_weeks_before_sustainable_packaging) as actual_sales_difference,
    100*(CAST(total_sales_12_weeks_after_sustainable_packaging as decimal) - total_sales_12_weeks_before_sustainable_packaging)/total_sales_12_weeks_before_sustainable_packaging
        as percentage_change
from sales_breakup;
```

	actual_sales_difference	percentage_change
1	-6020455870	-8.74635756700206

2018

```sql
with sales_breakup as (
    select
    SUM(
        CASE
            when week_date between DATEADD(WEEK,-12,'2018-06-15') and DATEADD(DAY,-1,'2018-06-15') then CAST(sales as bigint)
            else 0
        END
      ) as total_sales_12_weeks_before_sustainable_packaging,
    SUM(
        CASE
            when week_date between '2018-06-15' and DATEADD(WEEK,11,'2018-06-15') then CAST(sales as bigint)
            else 0
        END
      ) as total_sales_12_weeks_after_sustainable_packaging
    from weekly_salesss
)
select (total_sales_12_weeks_after_sustainable_packaging - total_sales_12_weeks_before_sustainable_packaging) as actual_sales_difference,
    100*(CAST(total_sales_12_weeks_after_sustainable_packaging as decimal) - total_sales_12_weeks_before_sustainable_packaging)/total_sales_12_weeks_before_sustainable_packaging
        as percentage_change
from sales_breakup;
```

	actual_sales_difference	percentage_change
1	-4487151690	-7.01494250759442

OHHHHH the users do concern them with sustainable packaging making our initial claims wrong because for the same before and after period for 2019 and 2018 the sales numbers are atrocious.

Which areas of the business have the highest negative impact in sales metrics performance in 2020 for the 12 week before and after period?

- `region`
- `platform`
- `age_band`
- `demographic`
- `customer_type`

```sql
with sales_breakup as (
    select region,
    SUM(
        CASE
            when week_date between DATEADD(WEEK,-12,'2018-06-15') and DATEADD(DAY,-1,'2018-06-15') then CAST(sales as bigint)
            else 0
        END
      ) as total_sales_12_weeks_before_sustainable_packaging,
    SUM(
        CASE
            when week_date between '2018-06-15' and DATEADD(WEEK,11,'2018-06-15') then CAST(sales as bigint)
            else 0
        END
      ) as total_sales_12_weeks_after_sustainable_packaging
    from weekly_salesss
    group by region
)
, percentage_chg as (
    select *,(total_sales_12_weeks_after_sustainable_packaging - total_sales_12_weeks_before_sustainable_packaging) as actual_sales_difference,
        100*(CAST(total_sales_12_weeks_after_sustainable_packaging as decimal) - total_sales_12_weeks_before_sustainable_packaging)/total_sales_12_weeks_before_sustainable_packaging
            as percentage_change
    from sales_breakup
)
select AVG(percentage_change) as AVG_percentage_change_in_sales from percentage_chg
```

REGION

	AVG_percentage_change_in_sales
1	-6.03064568611864

PLATFORM

	AVG_percentage_change_in_sales
1	-5.08714349510942

AGE_BAND

	AVG_percentage_change_in_sales
1	-7.26386574576412

DEMOGRAPHIC

	AVG_percentage_change_in_sales
1	-7.06153434404266

CUSTOMER_TYPE

	AVG_percentage_change_in_sales
1	-7.48746968410653

Customer type has negatively impacted the sales before and after for the 12 weeks (about 3 months) period.